Molecular Basis of Inheritance - 100 MCQsMolecular Basis of Inheritance - 100 MCQs

NCERT Nichod : Biology Practice Series NEET 2025

Anmol Sharma • Lesson 22 • Feb 1, 2025

Molecular Basis of Inheritance

BIOLOGYLIVE

Title: Molecular Basis of Inheritance

1. DNA polymorphism forms the basis of :

(1) DNA finger printing

(2) Both genetic mapping and DNA finger printing

(3) Translation

(4) Genetic mapping

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2. The ratio between purines and pyrimidines in the starting codon

(1) 2 : 1

(2) 1 : 1

(3) 1 : 2

(4) 3 : 1

3. The cellular factory / Organelle responsible for the synthesis of proteins is

(1) Ribosome

(2) Nucleus

(3) Golgi body

(4) Centriole

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4. A coding sequence made of alternating C and U bases would form a polypeptide having ____

(1) Either leu or ser residue

(2) Alterating leu and ser residues

(3) Only ser residues

(4) Only leu residues

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5. Presence of exons and introns is seen in

(1) Secondary transcript

(2) Primary transcript

(3) t RNA

(4) snRNA

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6. DNA-dependent RNA polymerase also catalyse the polymerisation in only one direction,

(1) $3' \rightarrow 3'$

(2) $5' \rightarrow 5'$

(3) $5' \rightarrow 3'$

(4) $3' \rightarrow 5'$

7. In eukaryotes, RNA polymerase III catalyses the synthesis of

(1) 5 srRNA, tRNA & SnRNA

(2) mRNA, HnRNA & SnRNA

(3) 28 S rRNA, 18 S rRNA & 5 S rRNA

(4) All types of rRNA & tRNA

8. Transcription and translation can be couple in

(1) Yeast

(2) Paramecium

(3) E.Coli

(4) Amoeba

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9. Discontinuously synthesized strands of DNA adjacent to____ DNA template are joined together by______.

(1) 5 → 3, DNA endonucleace

(2) 3′ → 5′ DNA Aase

(3) 5′ → 3′ DNA ligase

(4) 3′ → 5′, DNA ligase

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10. Transcription is intiated by

(1) Only σ (sigma) factor

(2) Only DNA dependent RNA polymerase

(3) Only ρ (Rho) factor

(4) RNA polymerase when it associates transiently with σ (sigma) factor

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11. In a crime investigation, the investigating officer collects different biological samples from the crime spot for DNA fingerprinting analysis. Which of the following samples is not helpful in this analysis?

(1) Skin Shreds

(2) Erythrocytes

(3) Semen sample

(4) Hair follicle

12. What is the role of RNA polymerase III in the process of transcription in Eukaryotes?

(1) Transcription of rRNAs (28S, 18S and 5.8S)

(2) Transcription of tRNA, 5S rRNA and snRNA

(3) Transcription of precursor of mRNA

(4) Transcription of only snRNAs

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13. Genetic material of bacteriophage $\phi \times 174$ and it has ____ nucleotides

(1) ss DNA, 5386

(2) ss RNA, 5386

(3) ds DNA,5368

(4) ss DNA, 5368

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14. In a 3.2 Kbp long piece of DNA, 820 adenine bases were found. What would be the number of cytosine bases?

(1) 1560

(2) 780

(3) 1480

(4) 740

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15. If a geneticist uses the blind approach for sequencing the whole genome of an organism, followed by assignment of function to different segments, the methodology adopted by him is called as :

(1) Gene mapping

(2) Expressed sequence tags

(3) Bioinformatics

(4) Sequence annotation

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16. Select the correct option

i. z gene codes → beta-galactosidase (β-gal)

ii. y gene codes → permease

iii. a gene → transacetylase

iv. i gene → repressor of the lac operon

(1) Only i is correct

(2) Only ii,iii is correct

(3) Only i, ii, iii is correct

(4) All are correct

17. If the sequence of bases in DNA is AATCCATG, then the sequence of bases in its transcript will be

(1) UUAGGUAC

(2) AAUCGAAU

(3) UUAGCUUA

(4) AAUCGAUG

18. The radioactive probe used by Taylor to detect distribution of newly synthesized DNA in chromosomes:

(1) Thymidine

(2) Adenosine

(3) Guanidine

(4) Cytosine

19. The year 1953 was known for the Double helical model of DNA given by_____ , also for discovery of ribosome by______, also for proposing the semi conservative DNA replication by_____ and also for the experiment done on chemical evolution by______

(1) Watson and Crick; G.Palade; Watson and Crick; Urey Miller

(2) Watson and Crick; G.Palade; Messelson and Stahl; Stanley Miller

(3) Watson and Crick; De Duve; Messelson and Stahl; Stanley Miller

(4) Watson and Crick; G.Palade; Watson and Crick; Stanley Miller

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20. UUU, UUC code for which amino acids ?

(1) Alanine, phenylalanine respectively

(2) Phenylalanine, alanine respectively

(3) Phenylalanine, methioninerespectively

(4) Phenylalanine, phenylalanine respectively

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21. For transcription the requirements are

i. DNA

ii.DNA dependent RNA polymerase

iii. Nuceloside triphosphate

iv. Helicase

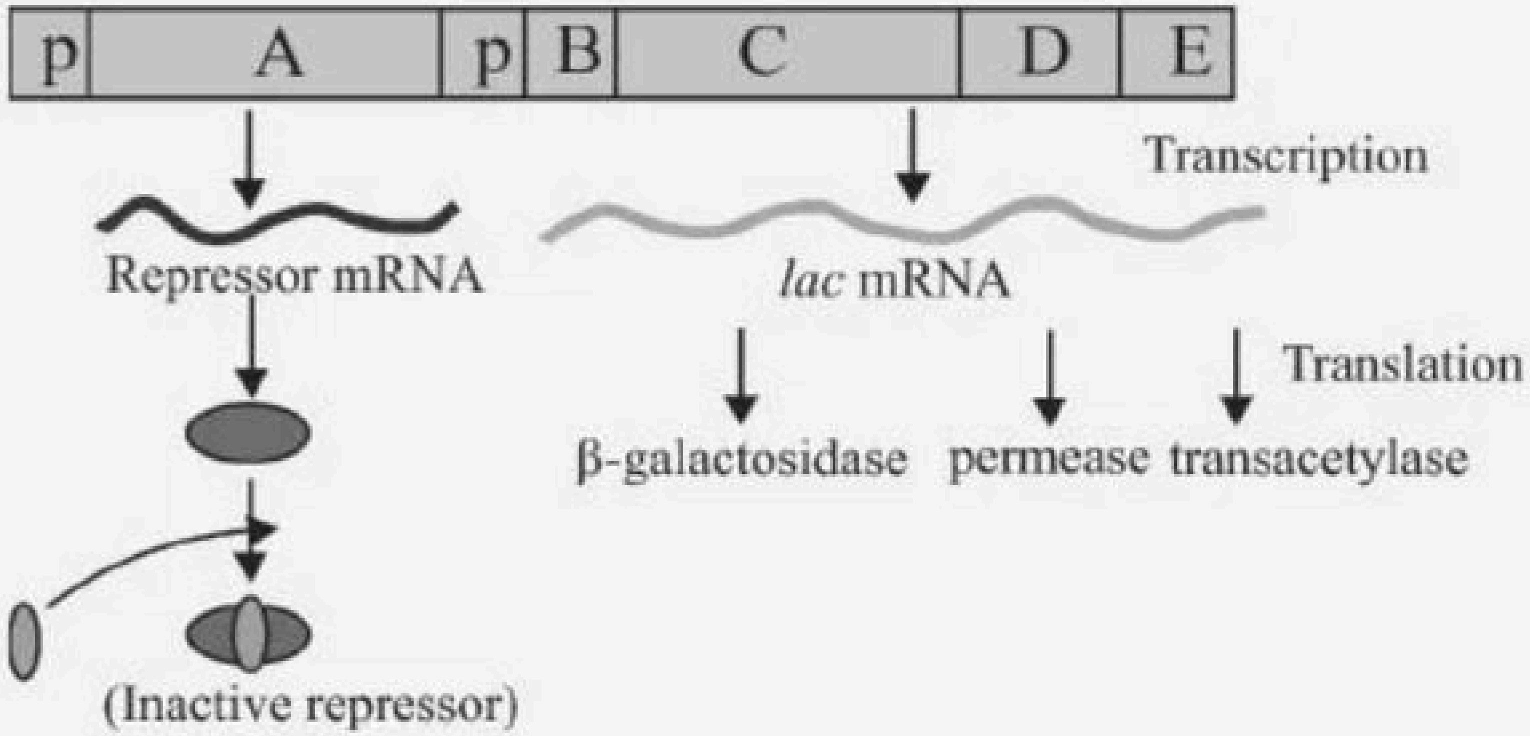
(1) i,ii,iii

(2) i,ii only

(3) i,ii,iv

(4) All

22. Identify A, B, C, D, E.



(1) A = Inhibitor B = Operator
C = y gene D = z gene
E = a gene

(2) A = inducer B = Operator
C = z gene D = y gene
E = a gene

(3) A = Inhibitor gene B = Operator
C = z gene D = y gene
E = a gene

(4) A = inducer B = Operator
C = y gene D = z gene
E = a gene

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23. During transcription the DNA strand with 3′ → 5′ polarity of the structural gene always acts as a template because

(1) Enzyme DNA dependent RNA polymerase always catalyse the polymerisation in 3′ → 5′ direction

(2) Enzyme DNA dependent RNA polymerase always catalyse polymerisation in both the directions

(3) Nucleotides of DNA strand with 5′ → 3′ are transferred to mRNA

(4) Enzyme DNA dependent RNA polymerase always catalyse the polymerisation in 5′ → 3′ direction

24. Select the correct statement

(1) Two strands of DNA are parallel and complimentary

(2) Two strands of DNA are anti- parallel and noncomplimentary

(3) Two strands are anti- parallel and complimentary

(4) All the above

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25. Haploid content of human DNA is ____

(1) 1.65 × 109 bP

(2) 3.1 – 3.3 × 109 bP

(3) 4.6 × 109 bP

(4) 3.3 × 109 bP

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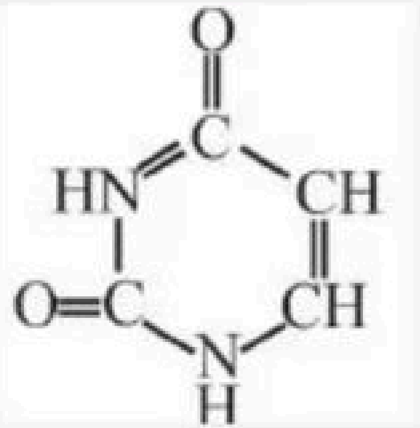
26. Identify the structure.



(1) Cytosine

(2) Thiamine

(3) Uracil

(4) Thymine

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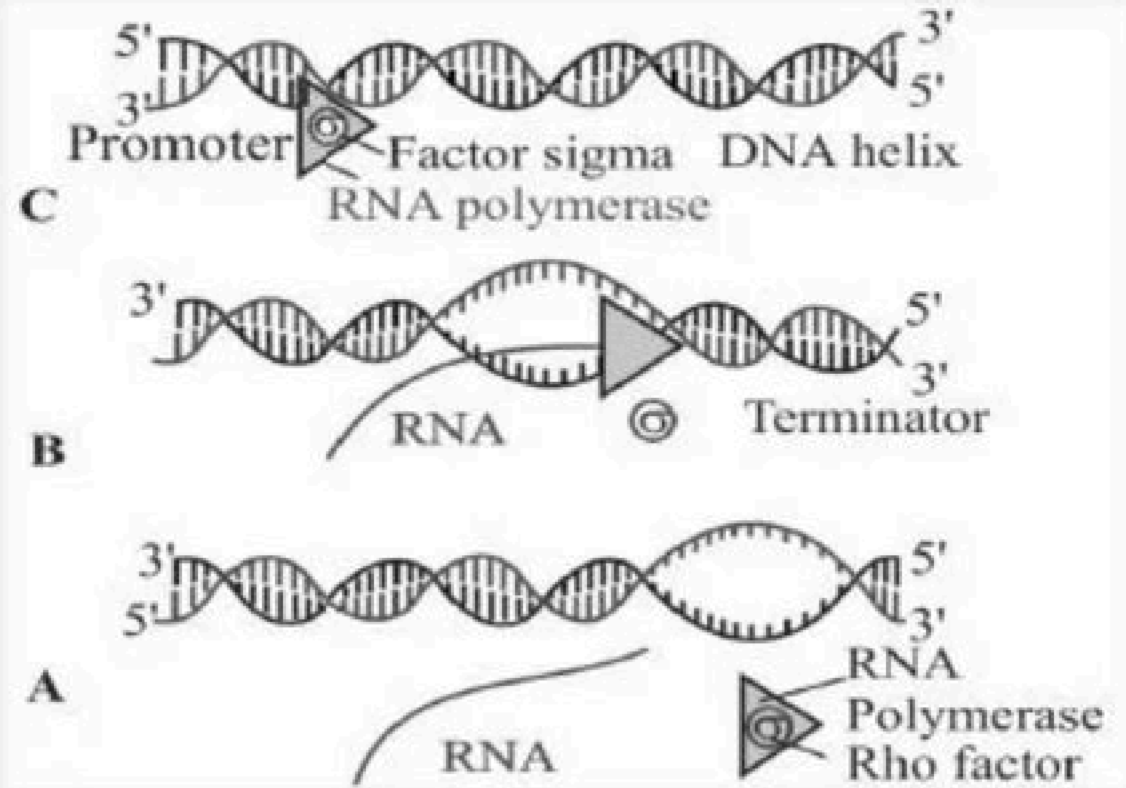
27. In the following identify the name of the process



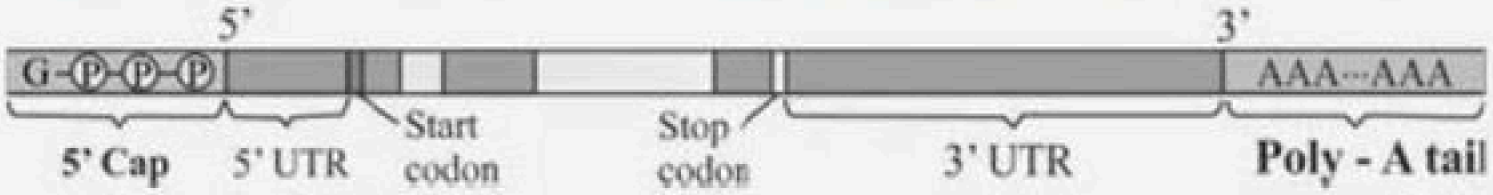
(1) A - Initiation, B - Elongation, C - Termination

(2) A - Termination, B - Elongation, C - Initiation

(3) A - Elongation, B - Initiation, C - Termination

(4) A - termination, B - Initiation, - Elongation

28. The following structure represents.



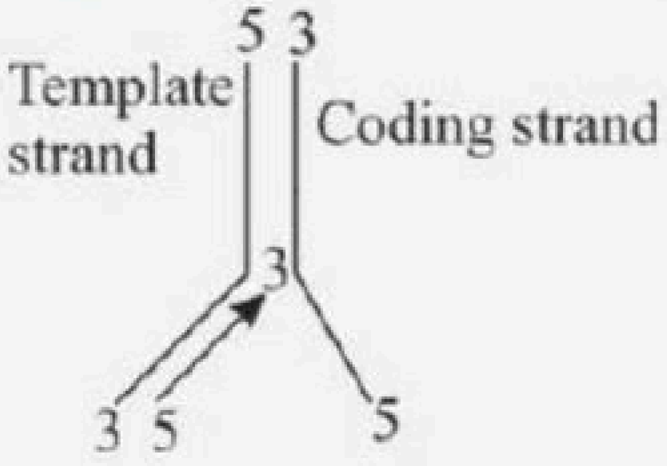
(1) r-RNA

(2) hn-RNA

(3) m-RNA

(4) t-RNA

29. This Diagram indicates



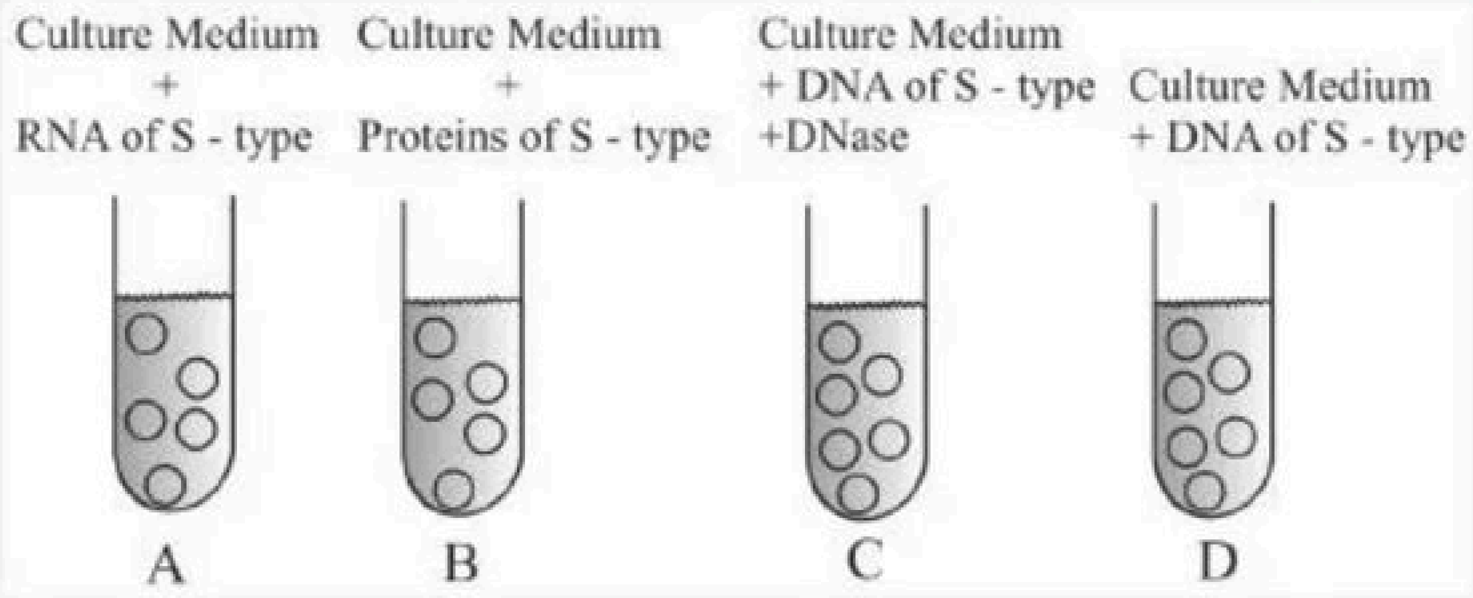
(1) DNA replication

(2) Reverse transcription

(3) Transcription

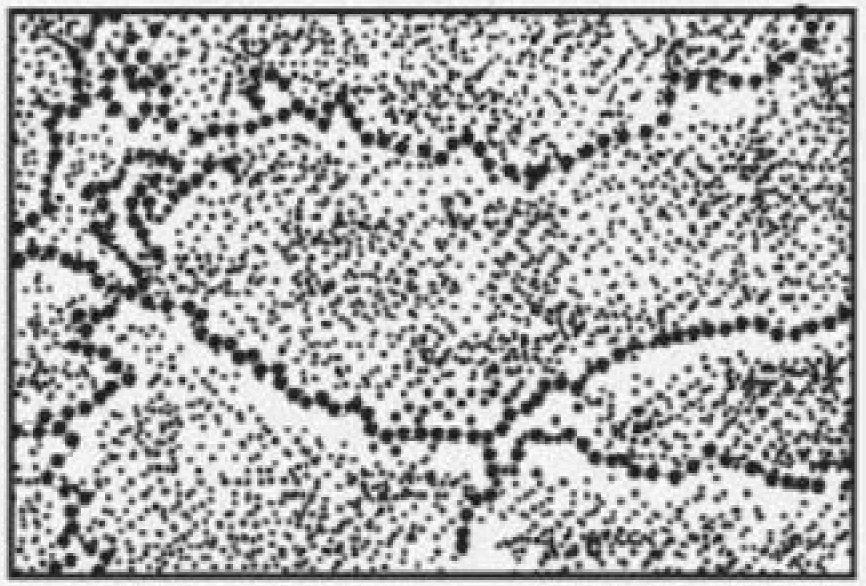
(4) Translation

30. Mention in which test-tube did Avery et al found transformation



(1) A
(2) B
(3) C
(4) D

31. The following structure represents



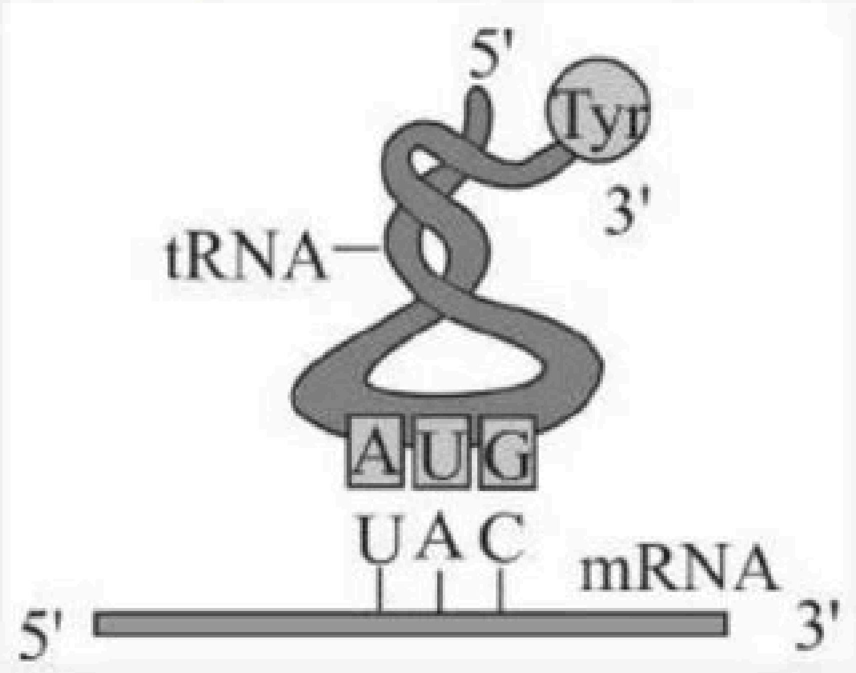
(1) 'Beads on string' structure of nucleosome in chromatin

(2) 'Beads on string' under electron microscope

(3) 'Beads on string' under compound microscope

(4) Both 1 & 2

32. In the following diagram the direction of anticodon is?



(1) t-RNA -3'-3'

(2) t-RNA-3'-5'

(3) t-RNA-5'-3'

(4) t-RNA-5'-5'

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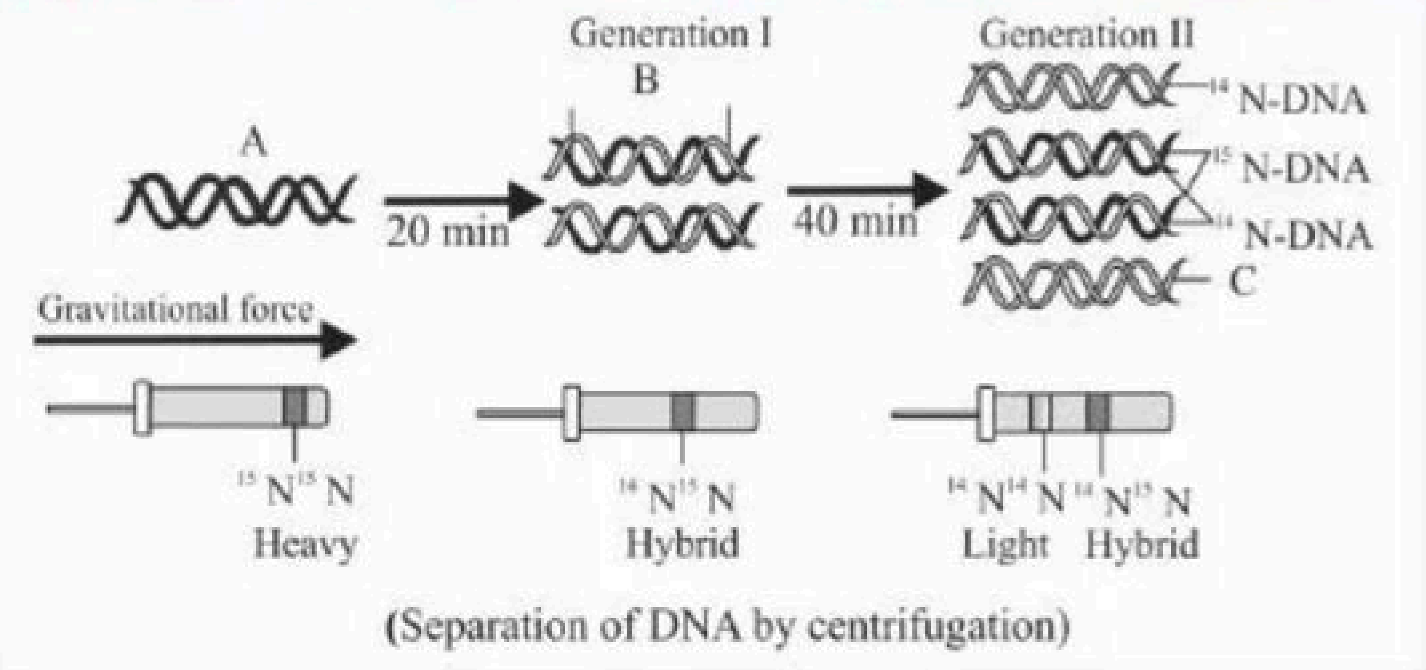
33. Which experiment is shown in the below figure?



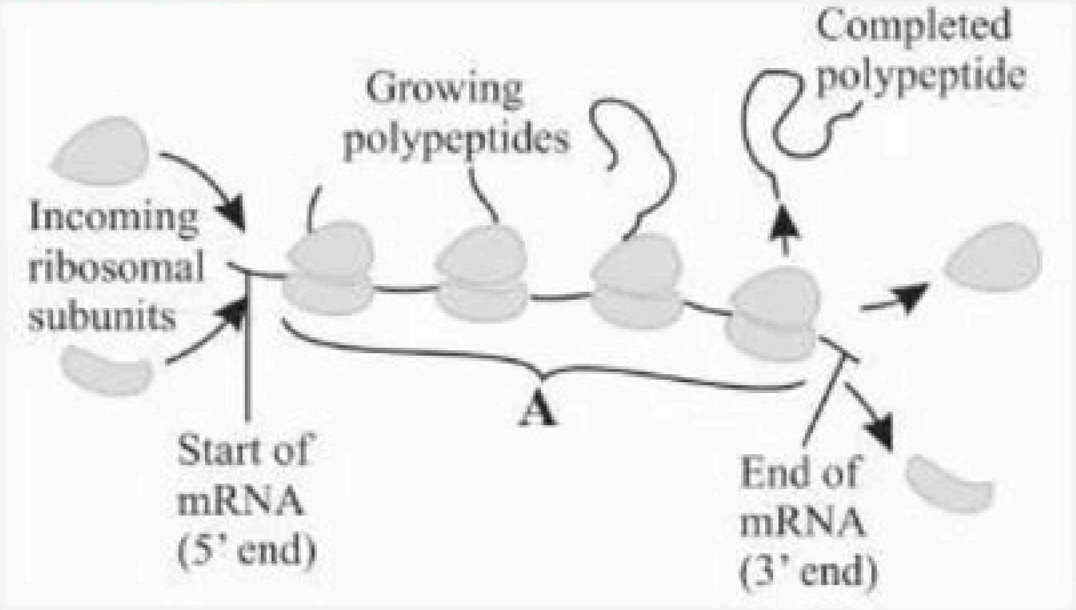
(1) Griffith's experiment (1928)

(2) Hershay and Chase experiment (1952)

(3) Meselson and Stahl experiment (1958)

(4) Watson and Crick experiment

34. The following represents a polysome, how many m- RNA are represented here ?



(1) Only one

(2) As many as polypeptides are present

(3) It can be variable

(4) None of these

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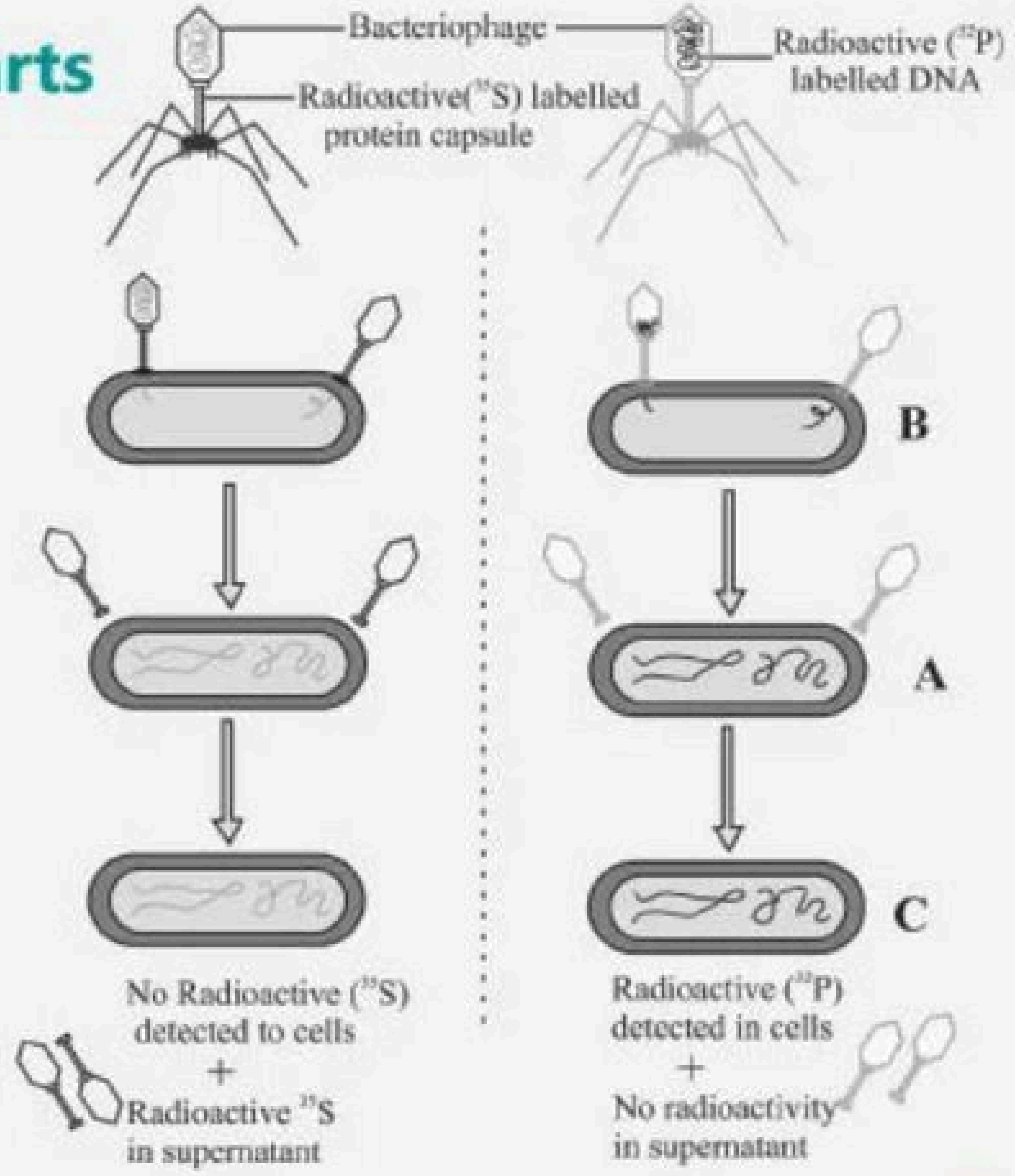
35. In the following experiment identify the labelled parts

(1) A - Infection, B - Blending, C - Centrifugation

(2) A - Centrifugation, B - Infection, C - Blending

(3) A - Blending, B - Infection, C - Centrifugation

(4) A - Centrifugation, B - Blending, C - Infection



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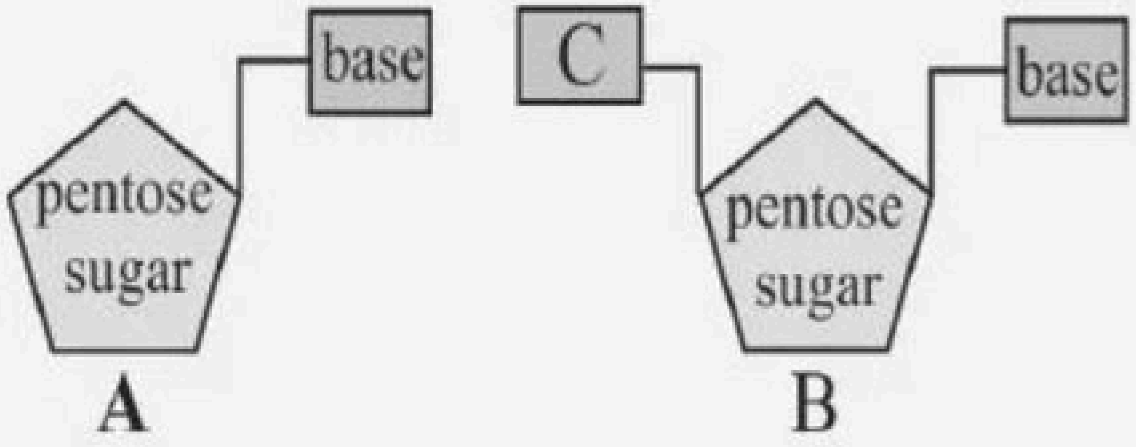
36. Choose the correct option for the below figure

(1) A - nucleotide; B - nucleoside; C - Phosphate

(2) A - nucleoside; B - nucleotide; C - Phosphate

(3) A - Phosphate; B - nucleotide; C - nucleoside

(4) A - nucleoside; B - Phosphate; C - nucleotide



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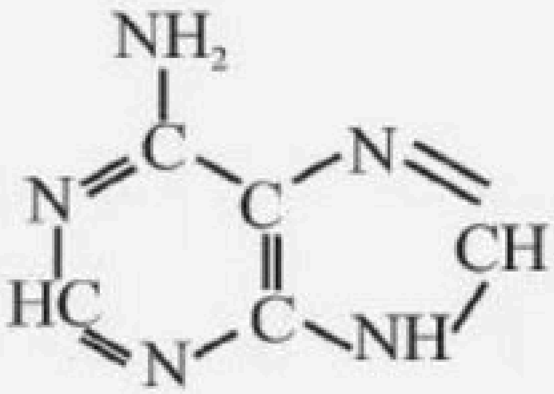
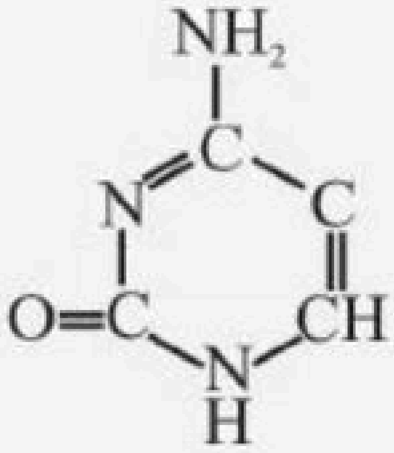
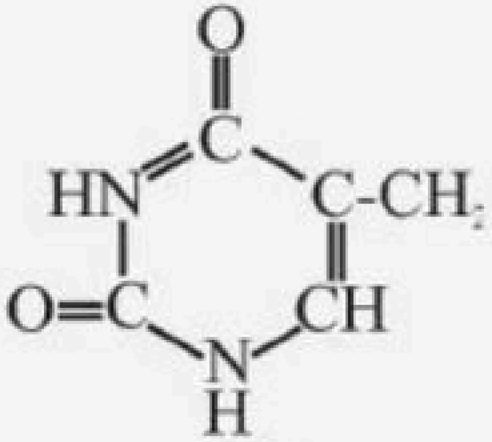
37. Identify the purine out of the structure.



(A)



(C)



(B)

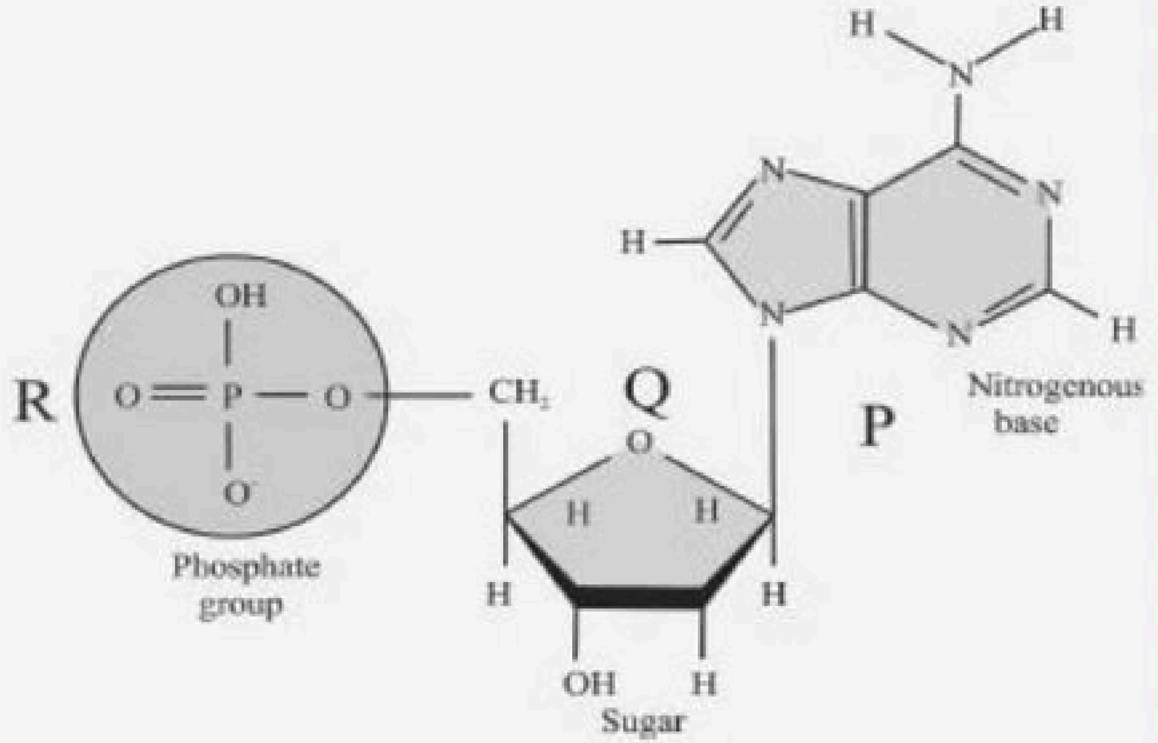
(1) B – Adenine

(2) A - Adenine

(3) C – Adenine

(4) A - Guanine

38. Identify .



(1) P - Phosphate group, Q-Deoxyribose sugar, RNitrogen base

(2) P-Nitrogen base, Q-Deoxyribose sugar, R-Phophate group

(3) P-Nitrogen base, Q-Ribose sugar, R-Phosphate group

(4) P-Phosphate group, Q-Ribose sugar, R-Nitrogen base

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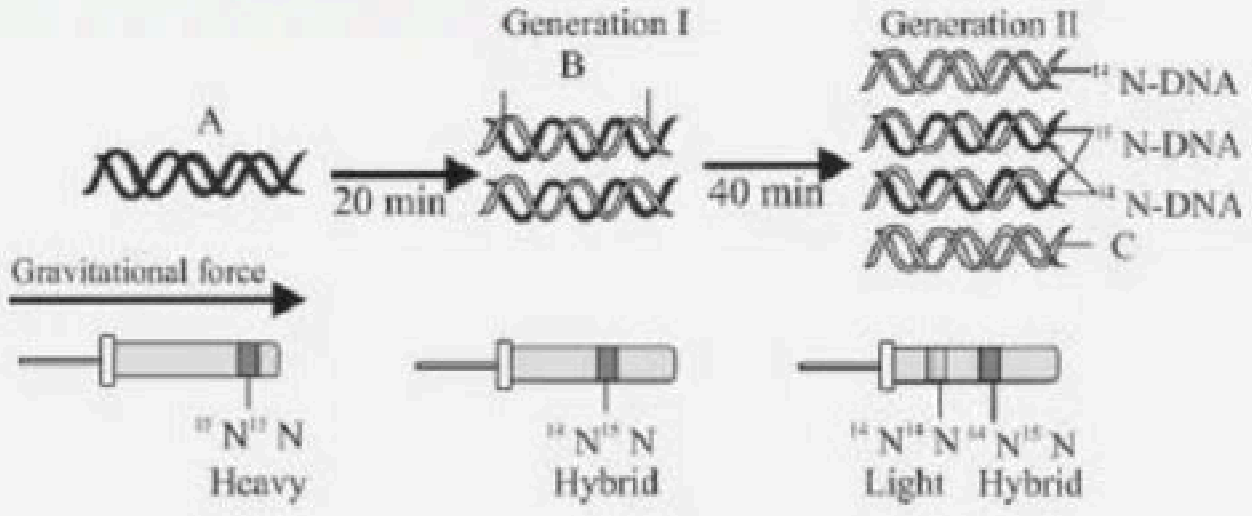
39. In the following diagram, depicting the Messelson & Stahl expt. Identify the type of DNA A, B & C.



(Separation of DNA by centrifugation)

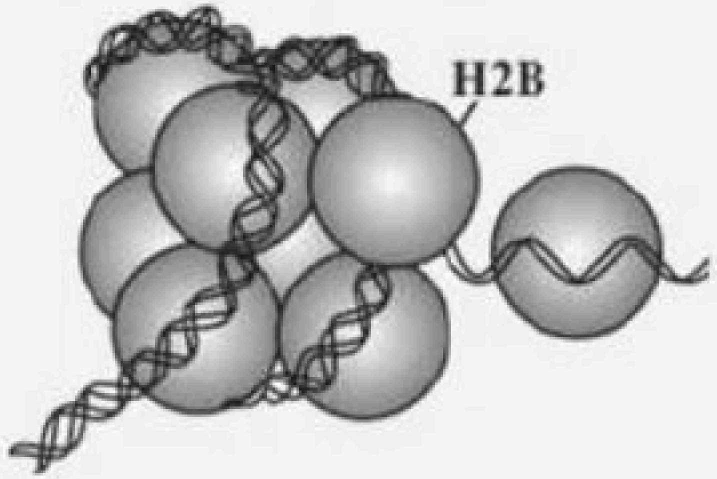
(1) A – $N^{14}\,N^{14}$, B – $N^{15}\,N^{14}$, C – $N^{15}\,N^{15}$

(2) A – $N^{15}\,N^{15}$, B – $N^{15}\,N^{14}$, C – $N^{14}\,N^{14}$

(3) A – $N^{14}\,N^{14}$, B – $N^{14}\,N^{14}$, C – $N^{15}\,N^{15}$

(4) A – $N^{15}\,N^{15}$, B – $N^{15}\,N^{15}$, C – $N^{14}\,N^{14}$

40. In the diagram how many histone proteins are depicted



(1) 9 (2) 8

(3) 10 (4) 11

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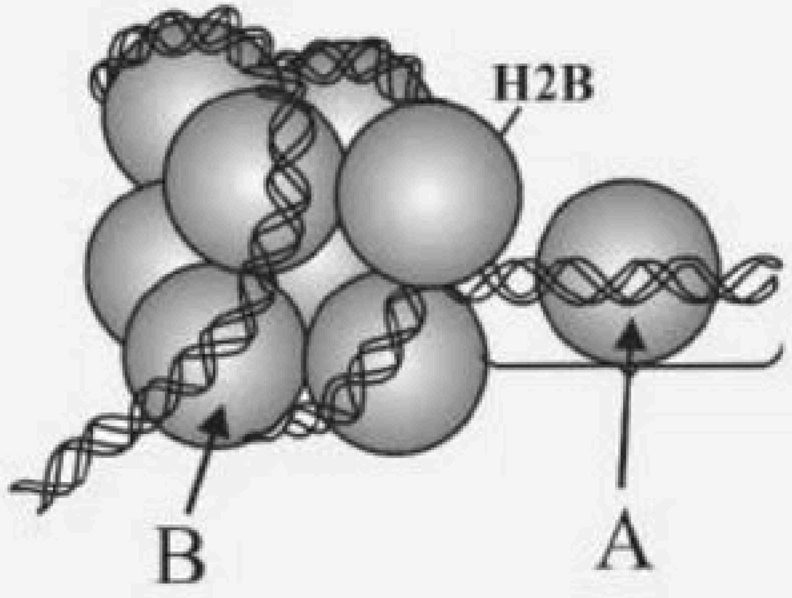
41. Identify the structure & labelled parts A, B.

(1) Nucleus – A – H1 B – Octamer

(2) Nucleus – A – Octamer B – H1

(3) Nucleosome A – H1 B – Octamer

(4) Nucleosome – A Octamer B – H1



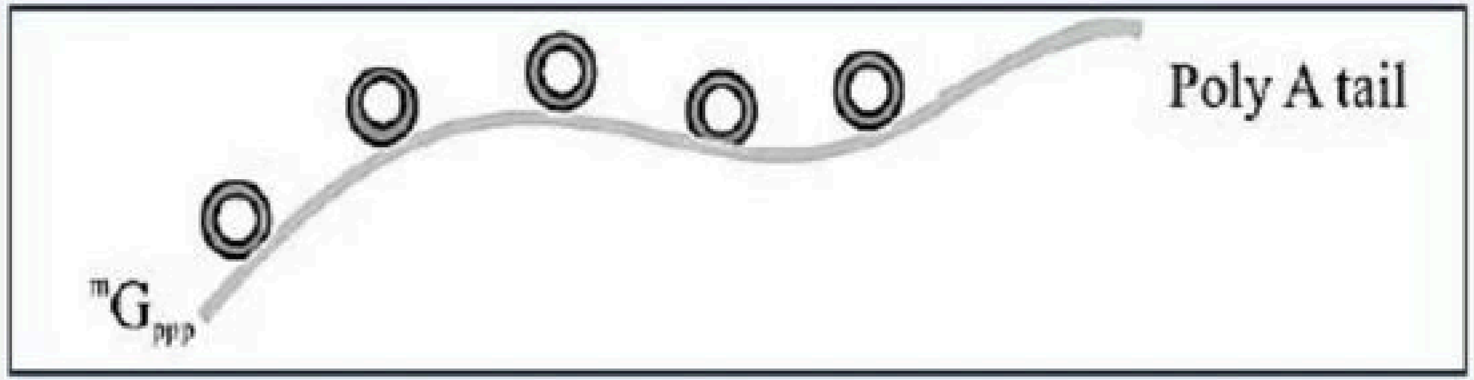
42. Identify the process showed in the process diagram

(1) Capping

(2) RNA splicing

(3) Poly adenylation

(4) All of these



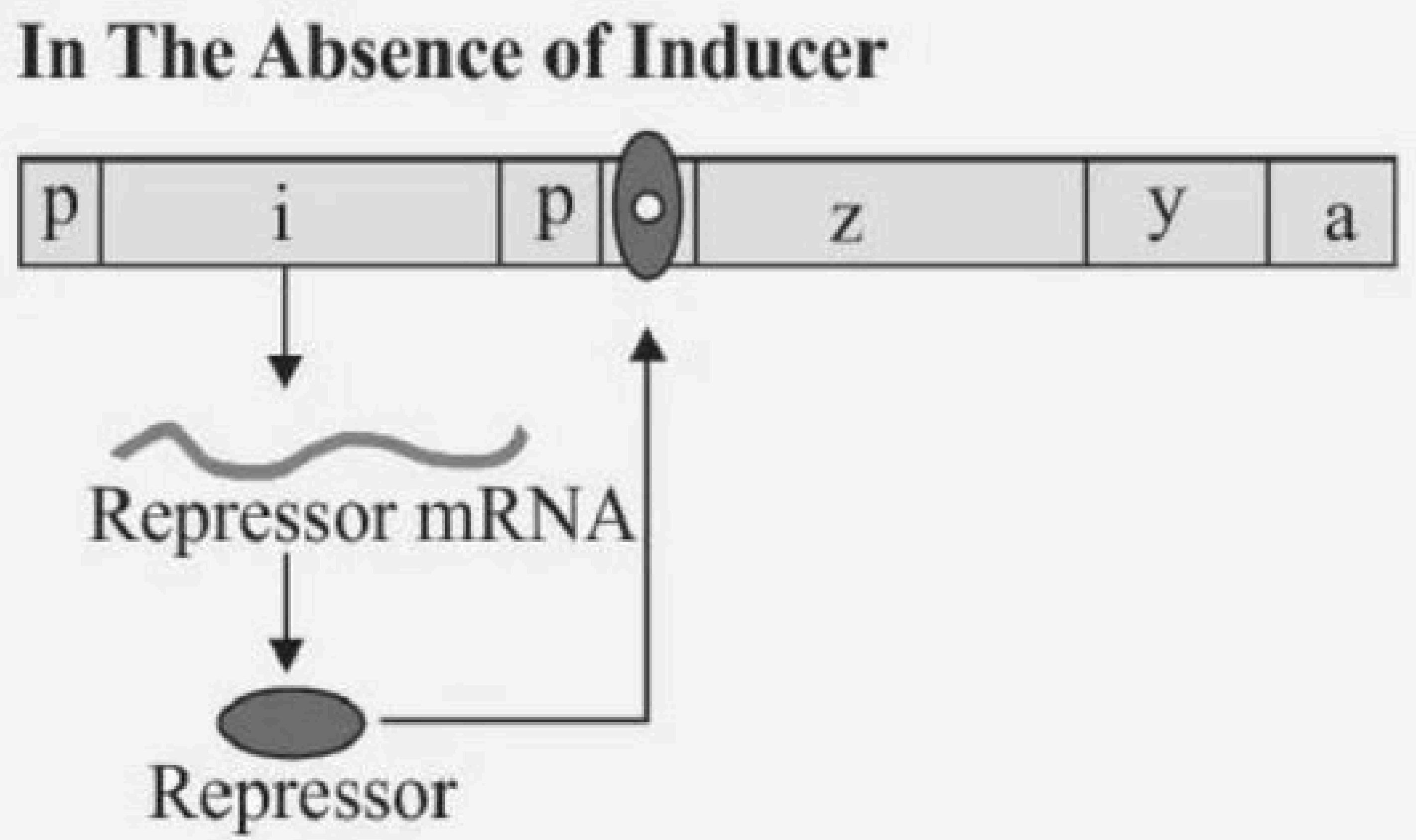
43. Identify the diagram.

(1) Lac operon (switched on)

(2) Lac operon (switched off)

(3) Try operon (switched on)

(4) Trp operon (switched off)



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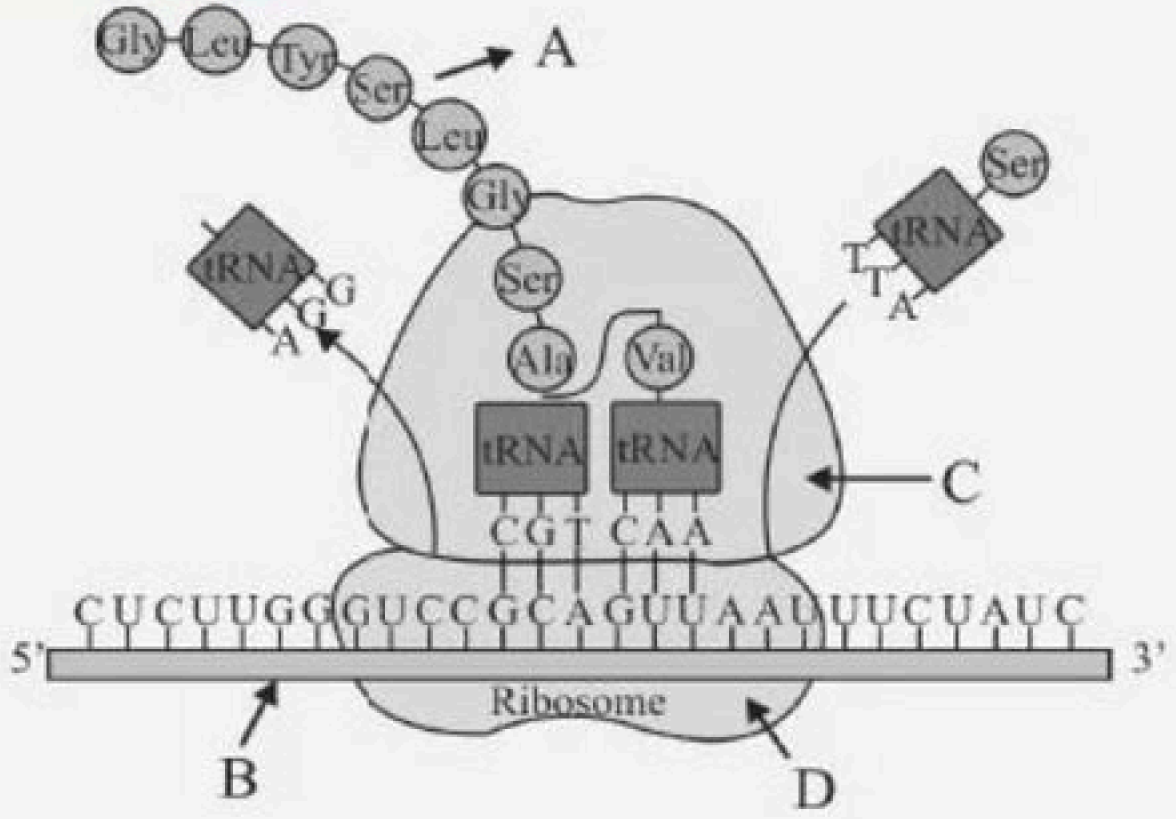
44. Identify the structure that contains codon (B) chain of amino acid, (A) large (C) small ribosomal unit (D) ribosomal unit in a prokaryote

(1) m-RNA, polypeptide, 50 S, 30 S

(2) r-RNA, polypeptide, 50 S, 30 S

(3) m-RNA, polypeptide, 60 S, 40 S

(4) m-RNA, polypeptide, 60 S, 40 S



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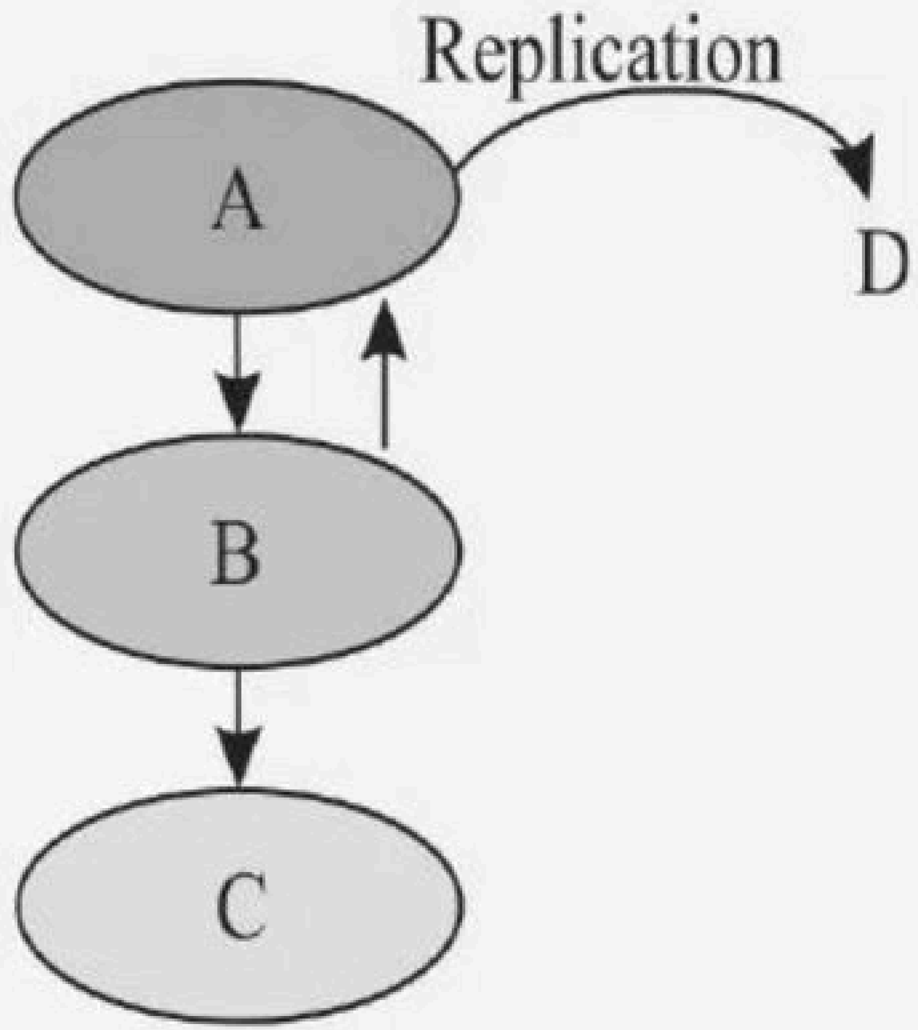
45. The diagram represents the central dogma of molecular biology. Choose the correct combination of labelling.



(1) A-DNA B-RNA C-Protein D-DNA

(2) A-RNA B-DNA C-Protein D-RNA

(3) A-DNA B-Protein C-Protein D-DNA

(4) A-Protein B-DNA C-DNA D-Protein

46. Match the columns I and II.

Column I		Column II	
A	DNA-dependent DNA polymerase	P	Replication is continuous
B	Deoxyribonucleoside triphosphates	Q	Average rate of polymerisation 2000 bp per second
C	Polarity $3' \rightarrow 5'$ of parental strand	R	Dual prupose
D	Polarity $5' \rightarrow 3'$ of oarental strand	S	Replication is discontinuous

(1) A – R, B – P, C – Q, D – S

(2) A – Q, B – R, C – P, D – S

(3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S

47. Match the columns I and II.

Column I		Column II	
A	Promoter	P	Provides binding site for DNA
B	Terminator	Q	Flanked portion
C	Structural gene	R	End of the process of transcription

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R

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48. Match the columns I and II WRT Translation.

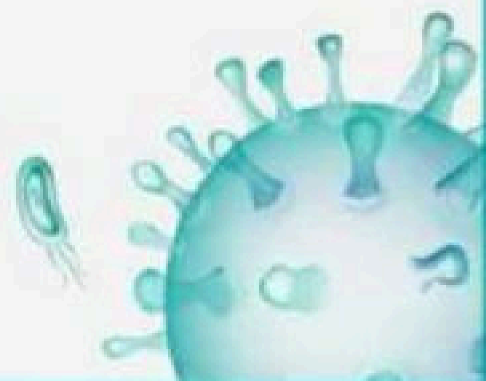
Column I		Column II	
A	Ribosome	P	Recognised only by the initiator tRNA
B	Release factor	Q	Moves from codon to codon
C	Start codon	R	Binds to the stop codon
D	UTRs	S	Present at the both 5' -end and 3' -end

(1) A – R, B – P, C – Q, D – S

(2) A – Q, B – R, C – P, D – S

(3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S



49. Match Column- I with Column- II

Choose the correct answer from the options given below:

Column I		Column II	
A	Gene 'a'	P	β-galactosidase
B	Gene 'y'	Q	Transacetylase
C	Gene 'i'	R	Permease
D	Gene 'z'	S	Repressor protein

(1) A-Q, B-P, C-S, D-R

(2) A-Q, B-R, C-S, D-P

(3) A-R, B-S, C-P, D-Q

(4) A-R, B-P, C-S, D-Q

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50. Match the columns I and II.

Column I		Column II	
A	Friedrich Meischer	P	Purine: pyrimidine ratio equal one
B	James Watson & Francis Crick	Q	Nuclein
C	Erwin Chargaff	R	1953

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R

51. Match the columns I and II wrt HGP.

Column I		Column II	
A	*Caenorhabditis elegans*	P	Plant
B	*Arabidopsis*	Q	Genes that are expressed as RNA
C	EST	R	A free living non-pathogenic nematode
D	Sequence annotation	S	Assigning different regions in the sequence with functions

(1) A – R, B – P, C – Q, D – S
(2) A – Q, B – R, C – P, D – S
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S

52. Match the columns I and II wrt HGP.

Column I		Column II	
A	Human genome	P	3000 bases
B	Average gene	Q	3164.7 million nucleotide
C	Chromosome I	R	2968 genes
D	Y chromosome	S	231 genes

(1) A – R, B – P, C – Q, D – S
(2) A – Q, B – R, C – P, D – S
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S

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53. Match the columns I and II.

Column I		Column II	
A	Exons	P	Absent in RNA
B	Introns	Q	Functional unit of inheritance
C	Gene	R	Processed RNA

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R

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54. Match the columns I and II with respect to DNA.

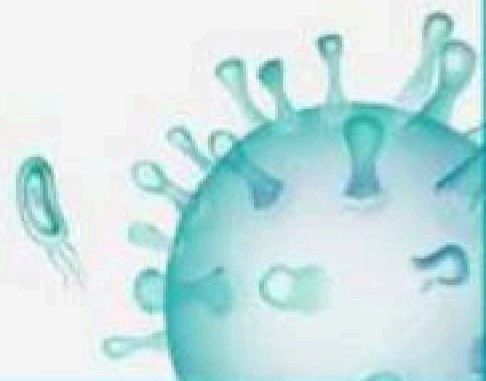
Column I	Column II
A Base pairing	P Sugar of DNA
B Deoxyribose	Q Unique property
C Thymine	R Nitrogen base of DNA

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R



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55. Match the columns I and II.

Column I	Column II
A Capping	P 5′ -end of hnRNA
B Splicing	Q 3′ -end of hnRNA
C Tailing	R Removal of introns

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R

56. Match the columns I and II.

Column I		Column II	
A	DNA-dependent DNA polymerase	P	Joining of discontinuously synthesized fragments
B	DNA ligase	Q	Polymerisation only in one direction $5' \rightarrow 3'$
C	Origin of replication	R	Initiate replication

(1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R

57. Match the columns I and II.

Column I		Column II	
A	Process of making RNA from DNA	P	Human genome project
B	Process of protein synthesis	Q	Transcription
C	Determination of complete nucleotide sequence	R	Translation

(1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R

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58. Match the columns I and II.

Column I		Column II	
A	Unambigous	P	Some amino acids are coded by more than one codon
B	Degenerate	Q	AUG
C	Dual functions	R	One codon codes for only one amino acid
D	Universal	S	From bacteria to human UUU would code for phenylalanine

(1) A – R, B – P, C – Q, D – S
(2) A – Q, B – R, C – P, D – S
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S

59. Match the columns I and II.

Column I		Column II	
A	AUG	P	Phenylalanine
B	UUU	Q	Methionine
C	UUC	R	Phenylalanine
D	UAG	S	Non-sense codon

(1) A – R , B – P , C – Q, D – S

(2) A – Q , B – R , C – S, D –P

(3) A – P , B – R , C – Q, D – S

(4) A – Q , B – P , C – R , D – S

60. Match the columns I and II.

Column I		Column II	
A	mRNAmessenger RNA	P	In ribosomes in cytoplasm
B	tRNATransfer RNA	Q	From nucleus to cytoplasm
C	rRNAribosomal RNA	R	In cytoplasm

(1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R

Title: Molecular Basis of Inheritance

61. The codon on mRNA are:

CAU-CCU-AAA-CUG.

Identify the correct sequence of amino acids.

(1) His-Pro-Lys-Leu

(2) Pro-His-Lys-Leu

(3) His-Pro-Leu-Lys

(4) Pro-Leu-Lys-His

Title: Molecular Basis of Inheritance

62. Match the columns I and II.

Column I		Column II	
A	Repressor	P	Increases permeability of lactose
B	Betagalactosidase	Q	Hydrolysis of the disaccharide
C	Permease	R	Acetylation of galactosides
D	Transacetylase	S	Binds to the operator

(1) A – R, B – P, C – Q, D – S
(2) A – S, B – Q, C – P, D – R
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S

Title: Molecular Basis of Inheritance

63. Match the columns I and II.

Column I		Column II	
A	Frederick Griffith	P	*Streptococcus pneumoniae*
B	Alfred Hershey & Matha Chase	Q	*E.coli*
C	Mathew Messelson & Franklin Stahl	R	Bacteriophages
D	Taylor	S	*Vicia faba*

(1) A – R, B – P, C – Q, D – S
(2) A – Q, B – R, C – P, D – S
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S

64. Match the columns I and II.

Column I		Column II	
A	mRNA messenger RNA	P	Structural & catalytic role
B	tRNA- Transfer RNA	Q	Template for translation
C	rRNA- ribosomal RNA	R	Brings aminoacids and reads the genetic code

(1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R

65. Match the columns I and II.

Column I		Column II	
A	Cistron	P	Eukaryotes
B	Monocistronic	Q	DNA coding for a polypeptide
C	Polycistronic	R	Bacteria

(1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R

66. Match the columns I and II.

Column I		Column II	
A	Promoter	P	5' -end (upstream) of the structural gene
B	Terminator	Q	Upstream or downstream to the promoter
C	Additional regulatory sequences	R	3' -end (downstream) of the coding strand

(1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R

Title: Molecular Basis of Inheritance

67. Match the columns I and II.

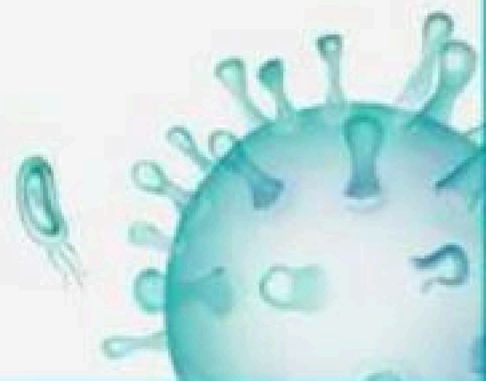
Column I		Column II	
A	Replication fork	P	Polyploidy
B	Replication of DNA	Q	Small opening of the DNA helix
C	Failure in cell division after DNA replication	R	S-phase of the cell cycle

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R



Title: Molecular Basis of Inheritance

68. Match the columns I and II wrt tRNA.

Column I		Column II	
A	Anticodon loop	P	It binds to specific amino acids
B	Amino acid acceptor end	Q	Clover-leaf
C	Secondary structure	R	Complementary
D	Actual structure	S	Inverted L

(1) A – R, B – P, C – Q, D – S
(2) A – Q, B – R, C – P, D – S
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S

Title: Molecular Basis of Inheritance

69. Match the columns I and II.

Column I		Column II	
A	N-glycosidic linkage	P	Linkage of two nucleotides
B	Phosphoester linkage	Q	Nitrogenous base is linked to the C-1 of pentose sugar
C	Phosphodiester bonds	R	Phosphate group is linked to the OH group of the 5-C
D	2 Hydrogen bonds	S	Between C & G
E	3 Hydrogen bonds	T	Between A & T

(1) A – R, B – P, C – Q, D – T, E – S
(2) A – Q, B – R, C – P, D – T, E – S
(3) A – S, B – R, C – P, D – T, E – Q
(4) A – Q, B – T, C – P, D – R, E – S

70. Match the scientists of Column I with their contributions in Column II

Column I		Column II	
A	Griffith	P	Lac operon
B	Jacob & Monad	Q	DNA is the genetic material
C	Meselson & Stahl	R	Transforming principle
D	Hershey and Chase	S	DNA replicates semi-conservativelys.

(1) A - P; B - Q; C - R; D - S
(2) A - P; B - S; C - Q; D - R
(3) A - R; B - P; C - S; D - Q
(4) A - R; B - Q; C - P; D - S

Title: Molecular Basis of Inheritance

71. Match the columns I and II with respect to nucleosome.

Column I		Column II	
A	Typical nucleosome	P	Negatively charged
B	Histones	Q	200 bp
C	DNA	R	Basic amino acid lysine and arginines

(1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R

Title: Molecular Basis of Inheritance

72. Match the columns I and II.

Column I		Column II	
A	Bacteriophage $\phi \times 174$	P	5386 nucleotides
B	Bacteriophage lambda	Q	4.6×10^6bp
C	*Escherichia coli*	R	48502 base pairs
D	Human DNA haploid	S	3.3×10^9 bp

(1) A – R, B – P, C – Q, D – S
(2) A – Q, B – R, C – P, D – S
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S

73. Match the columns I and II.

Column I		Column II	
A	Frederick Griffith	P	1952
B	Oswald Avery	Q	British bacteriologist - 1928
C	Alfred Hershey & Matha Chase	R	Radioactive thymidine
D	Taylor	S	1958
E	Mathew Messelson & Franklin Stahl	T	1933 - 34

(1) A – R, B – P, C – Q, D – T, E – S
(2) A – Q, B – R, C – P, D – T, E – S
(3) A – S, B – R, C – P, D – T, E – Q
(4) A – Q, B – T, C – P, D – R, E – S

74. Match the following:

Column I		Column II	
A	t-RNA	1	Linking of amino-acids
B	m-RNA	2	Transfer of genetic information
C	r-RNA	3	Nucleolar organising region
D	Peptidyl	4	Transfer of amino acid from transferase cytoplasm of ribosome

(1) A-4,B-2,C-3,D-1

(2) A-1,B-4,C-3, D-2

(3) A-1,B-2,C-3, D-4

(4) A-1,B-3,C-2,D-4

Title: Molecular Basis of Inheritance

75. Match the columns I and II.

Column I		Column II	
A	RNA polymerase I	P	Transcribes snRNAs
B	RNA polymerase II	Q	Transcribes rRNAs
C	RNA polymerase III	R	Transcribes hnRNA

(1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R

76. Match the following.

Column I		Column II	
A	VNTR	P	Largest gene
B	Introns and exons	Q	DNA fingerprinting
C	Dystrophin	R	Bulk DNA
D	Satellite DNA	S	Splicing

(1) A - Q; B - S;C - P;D - R

(2) A - S; B - P;C - Q;D - R

(3) A - R;B - S;C - P;D - Q

(4) A - Q;B - P;C - S;D - R

77. Match the columns I and II.

Column I		Column II	
A	Taylor & colleagues	P	Protein coat
B	Cesium chloride density gradient centrifugation	Q	1958, Visia faba
C	Radioactive sulphur (^{35}S)	R	Semi-conservative replication of DNA

(1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R

78. Match Column I with Column II

Choose the correct answer from the options given below:

Column I		Column II	
A	Frederick Griffith	P	Genetic code
B	Francois Jacob & Jacque Monod	Q	Semi-conservative mode of DNA replicatione
C	Har Gobind Khorana	R	Transformation
D	Meselson & Stahl	S	Lac operon

(1) A - R, B - Q, C - P, D - S

(2) A - R, B - S, C - P, D - Q

(3) A - Q, B - R, C - S, D - P

(4) A - S, B - P, C - Q, D - R

79. Match the columns I and II.

Column I		Column II	
A	*E.coli* DNA lengths	P	1.36 mm
B	Human DNA length	Q	1 μm
C	*E.coli* area	R	2.2 m
D	Human nucleus area	S	5 μm

(1) A – R, B – P, C – Q, D – S

(2) A – Q, B – R, C – P, D – S

(3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S

Title: Molecular Basis of Inheritance

80. Match the columns I and II.

Column I		Column II	
A	Non-histone chromosomal (NHC) proteins	P	Loosely packed chromatin
B	Euchromatin	Q	Higher level packaging of chromatin
C	Heterochromatin	R	Transcriptionally inactive

(1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R

Title: Molecular Basis of Inheritance

81. Match Column I with Column II:

Choose the correct answer from the options given below :

	Column I		Column II
A	RNA polymerase III	P	snRNPs
B	Termination of transcription	Q	Promotor
C	Splicing of Exons	R	Rho factor
D	TATA box	S	SnRNAs, tRNA

(1) A - Q, B - S, C - P, D - R
(2) A - R, B - Q, C - S, D - P
(3) A - R, B - S, C - P, D - Q
(4) A - S, B - R, C - P, D - Q

82. Match the columns I and II

Column I		Column II	
A	i gene	P	Transacetylase
B	z gene	Q	Repressor of the lac operon
C	y gene	R	Permease
D	a gene	S	Betagalactosidase

(1) A – R, B – P, C – Q, D – S
(2) A – Q, B – R, C – P, D – S
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – S, C – R, D – P

83. Match the columns I and II.

Column I		Column II	
A	H1 Histones	P	Inner core of nucleosom
B	H3 Histones	Q	Connects two nucleosome
C	Linker DNA	R	Present once per 200 base pairs
D	Human nucleus area	S	5 μm

(1) A – R, B – P, C – Q, D – S
(2) A – Q, B – R, C – P, D – S
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S

84. Match the columns I and II.

Column I		Column II	
A	RNA	P	Expression
B	Mendelian characters	Q	Nitrogen base of RNA
C	Uracil	R	QB bacteriophage

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – P,B – R, C – Q

(4) A – Q, B – P, C – R

Title: Molecular Basis of Inheritance

85. Match the columns I and II.

Column I		Column II	
A	George Gamow	P	Physicist
B	Har Gobind Khorana	Q	Cell-free system for protein synthesis
C	Marshal Nirenberg	R	Homopolymers & copolymers
D	Severo Ochoa	S	Polynucleotide phosphorylase

(1) A – R, B – P, C – Q, D – S
(2) A – Q, B – R, C – P, D – S
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S

86. Match the columns I and II.

Column I		Column II	
A	Friedrich Meischer	P	X-ray diffraction data
B	Maurice Wilkins & Rosalind Franklin	Q	Thymine
C	5-methyl uracil	R	1869

(1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R

87. Match the columns I and II wrt HGP.

Column I		Column II	
A	1990	P	Development of a new area in biology
B	2003	Q	Address the ethical, legal and social issues
C	ELSI	R	HGP was completed
D	Bioinformatics	S	HGP was launched

(1) A – R, B – P, C – Q, D – S
(2) A – Q, B – R, C – P, D – S
(3) A – S, B – R, C – Q, D – P
(4) A – Q, B – P, C – R, D – S

Title: Molecular Basis of Inheritance

88. Match the processes of Column I with the enzyme involved in Column II and select the correct options among the following

Select the code for the correct answer from the options given below

Column I		Column II	
A	DNA replication	P	RNA polymerase
B	Translation	Q	DNA polymerase
C	Transcription	R	Reverse transcriptase
D	Reverse transcription	S	Aminoacyl synthetase

(1) A - Q; B - S; C - R; D - P

(2) A - Q; B - S; C - P; D - R

(3) A - Q; B - R; C - S; D - P

(4) A - Q; B - S; C - R; D - P

Title: Molecular Basis of Inheritance

89. Statement A : DNA as an acidic substance present in nucleus was first identified by Friedrich Meischer in 1869.

Statement B : The base pairing confers a very unique property to the polynucleotide chains.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

Title: Molecular Basis of Inheritance

90. Matthew Meselson and Franklin Stahl grew____ in a medium containing $^{15}NH_4Cl$ (^{15}N is the __ isotope of nitrogen) as the only nitrogen source for many generations. The result was that_____ was incorporated into newly synthesised DNA (as well as other nitrogen containing compounds).

(1) E. coli, heavy, ^{15}N

(2) bacteriophage, heavy, ^{15}N

(3) E. coli, radioactive, ^{15}N

(4) bacteriophage, radioactive, ^{15}N

91. Sequences prior to the start point of transcription are called

(1) Non coding sequences

(2) Upstream sequences

(3) Coding sequences

(4) Downstream sequences

Title: Molecular Basis of Inheritance

92. Statement A : Griffith concluded that the strain bacteria had somehow been transformed by the heatkilled R strain bacteria.

Statement B : Some 'transforming principle', transferred from the heatkilled S strain, had enabled the R strain to synthesise a smooth polypeptide coat and become virulent.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

Title: Molecular Basis of Inheritance

93. Statement A : A single-strand of mRNA is capable of forming different polypeptide chains in prokaryotes.

Statement B : Termination codons occur in polypeptide.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

Title: Molecular Basis of Inheritance

94. The single stranded DNA of phage $\phi \times 174$ has 5400 nucleotides. If average protein contains 200 amino acids, how many different proteins could this phage DNA code ?

(1) 100

(2) 90

(3) 9

(4) 27

Title: Molecular Basis of Inheritance

95. In prokaryotes, DNA is held together by which proteins?

(1) Negatively charged proteins

(2) Non-histone chromosomal proteins

(3) Positively charged proteins

(4) Histone proteins

Title: Molecular Basis of Inheritance

96. If both the strands copied during transcription, then what will happen?

(1) The segment of DNA would be coding for two different proteins

(2) Two RNA will be produced simultaneously complementory to each other.

(3) There will be formation of double helical RNA

(4) All of the above

Title: Molecular Basis of Inheritance

97. Statement A : The unequivocal proof that DNA is the genetic material came from the experiments of Alfred Hershey and Martha Chase (1952).

Statement B : They worked with viruses that infect bacteria called bacteriophages.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

98. Name the nucleotide added to 5' end of hnRNA in capping.

(1) Ethyl guanosine monophosphate

(2) Ethyl guanosine triphosphate

(3) Methyl guanosine triphosphate

(4) Methyl guanosine monophosphate

Title: Molecular Basis of Inheritance

99. When mRNA is synthesized on DNA, the unwanted mRNA regions are removed and regions coding for amino acids are joined together, this process is called as

(1) Splicing

(2) Lysogeny

(3) Replication

(4) Central dogma

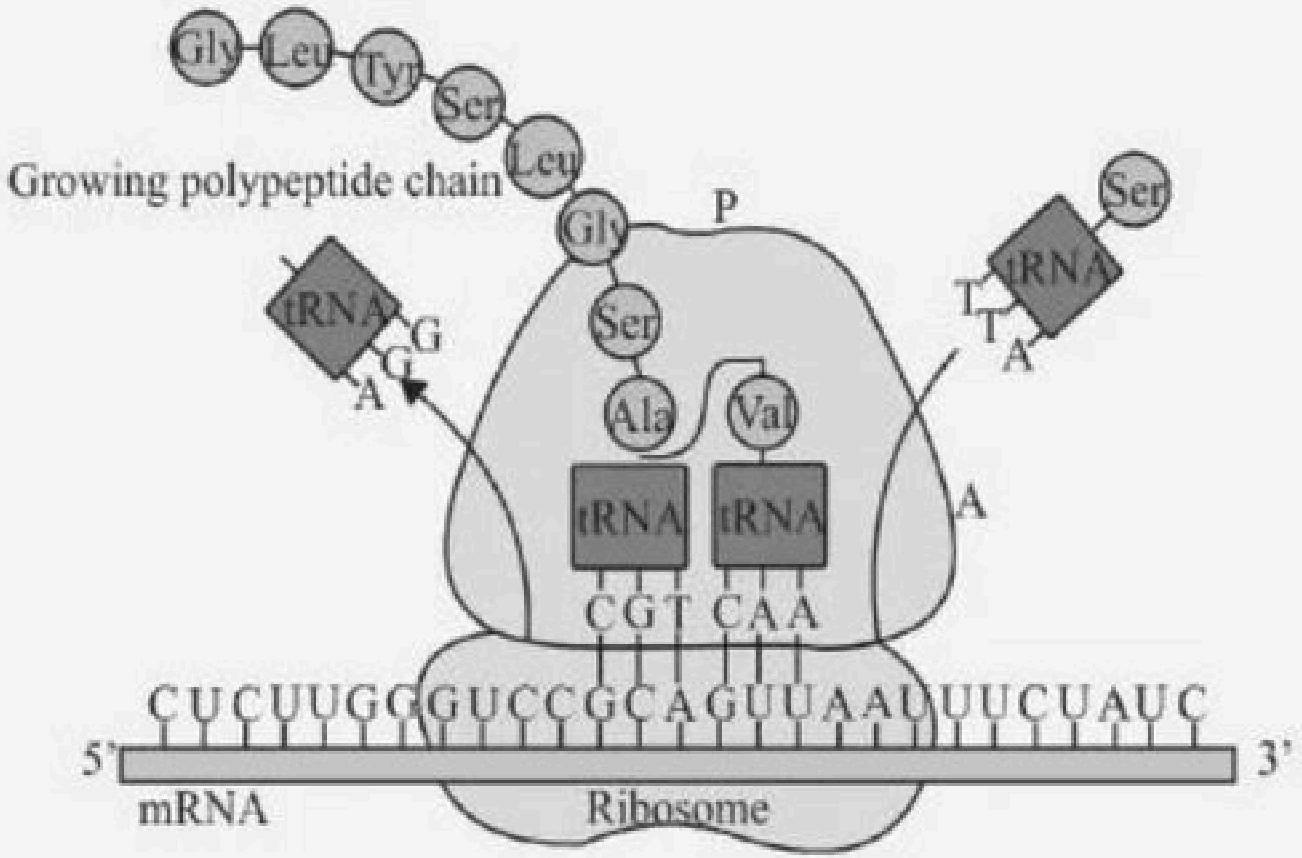
100. With respect to ribosome what does A & P represent.

(1) A - Peptidyl site P - t-RNA binding

(2) A - tRNA-binding site P - peptidyl site

(3) A - m - RNA site P - peptidyle site

(4) A - t-RNA binding site P - ATP site



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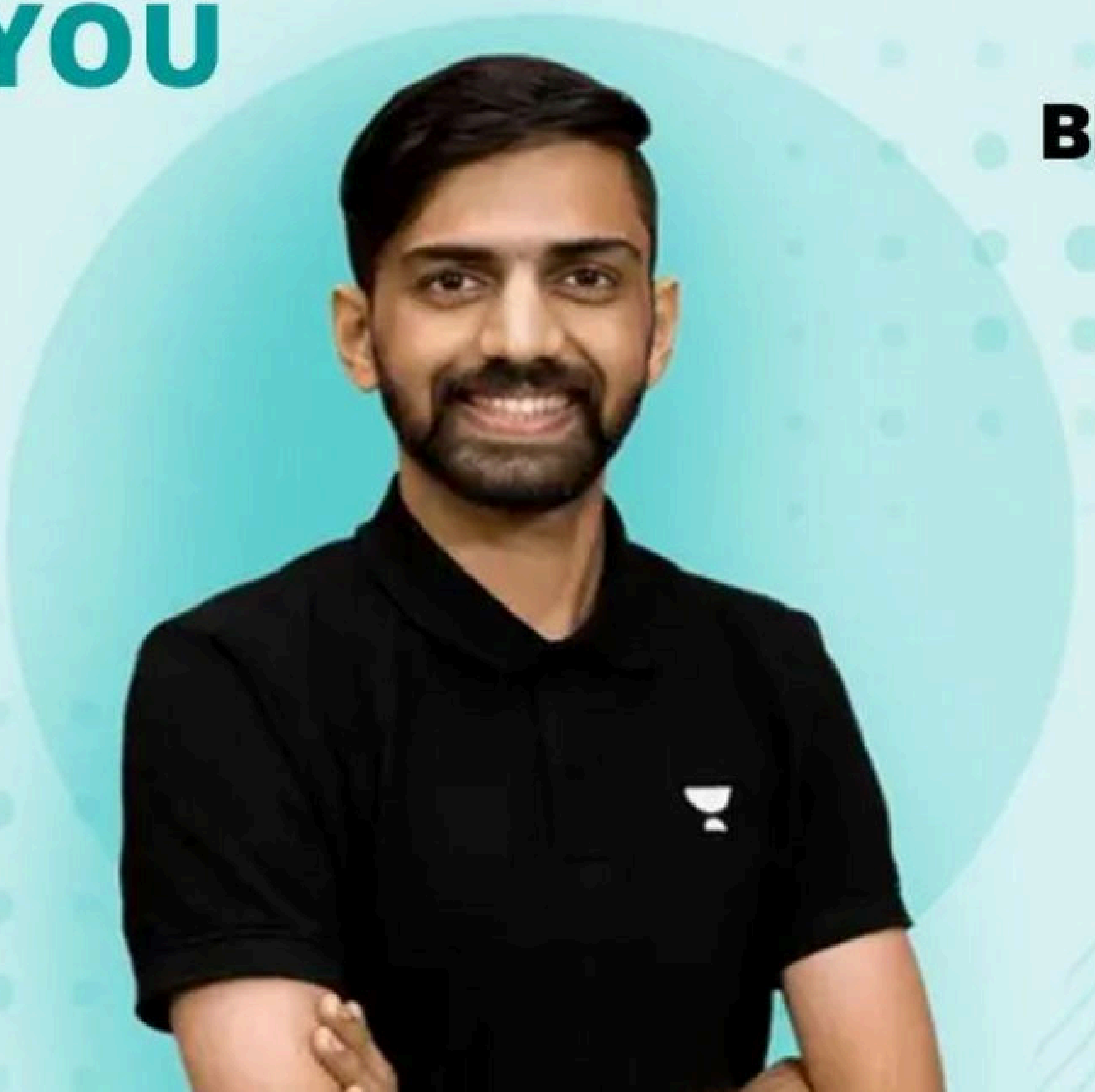

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